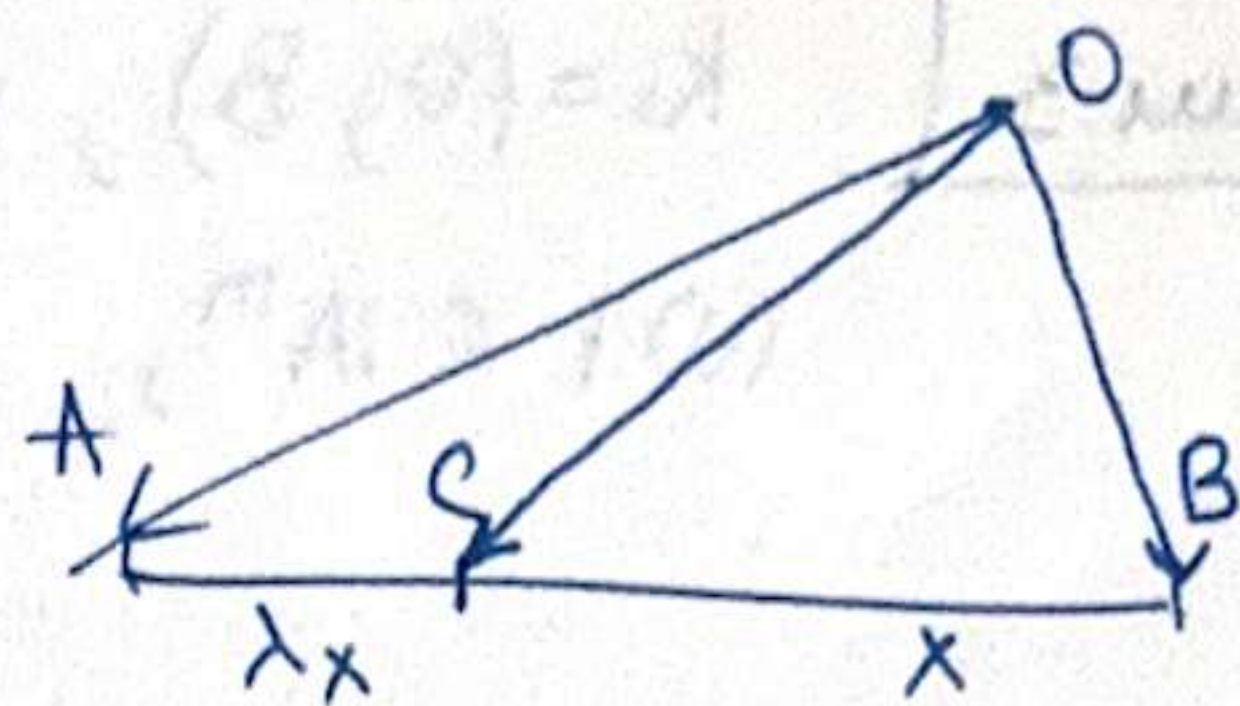


Theoretical Problems

Problem 1

$$\lambda = |CA| : |CB|$$



$$\vec{OC} = \frac{1}{1+\lambda} \vec{OA} + \frac{\lambda}{1+\lambda} \vec{OB} \quad ?$$

• Let $A(a_1, \dots, a_m)$

$B(b_1, \dots, b_m)$

• $A-C-B$ collinear. $\Rightarrow \vec{AC} = \lambda \cdot \vec{CB}$

$$\lambda = |CA| : |CB| \quad \Rightarrow \quad \vec{OC} - \vec{OA} = \lambda (\vec{OB} - \vec{OC})$$

$$\Rightarrow \vec{OC} + \lambda \cdot \vec{OC} = \vec{OA} + \lambda \cdot \vec{OB}$$

$$\Rightarrow (1+\lambda) \vec{OC} = \vec{OA} + \lambda \cdot \vec{OB} \quad | : (1+\lambda)$$

$$\Rightarrow \vec{OC} = \frac{1}{1+\lambda} \vec{OA} + \frac{\lambda}{1+\lambda} \vec{OB}$$

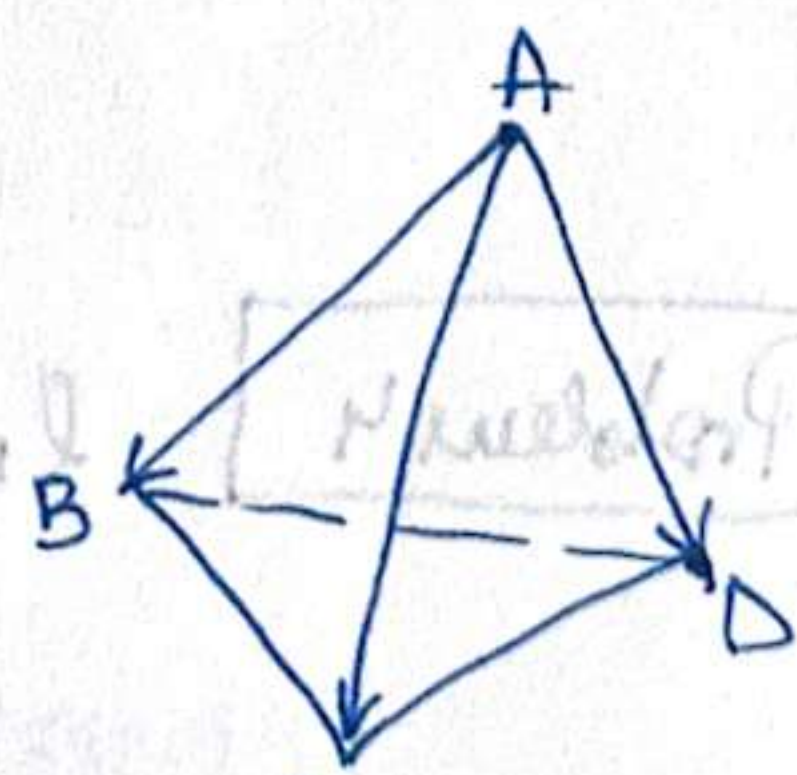
$$\Rightarrow C = \frac{1}{1+\lambda} (A + \lambda B) = \frac{A + \lambda B}{1+\lambda}$$

$$\Rightarrow \begin{cases} c_1 = \frac{a_1 + \lambda \cdot b_1}{\lambda + 1} \\ c_2 = \frac{a_2 + \lambda \cdot b_2}{\lambda + 1} \\ \vdots \\ c_m = \frac{a_m + \lambda \cdot b_m}{\lambda + 1} \end{cases}$$

Problem 2

How many Cartesian frames can we construct with the vertices of a tetrahedron?

A frame in $\mathbb{E}^3$ is: $K(o, B)$, $B = (i, j, k)$ basis. $\downarrow$ point



ABCD tetrahedron $\Rightarrow$ origin can be chosen in 4 ways:

Let A be the origin $\Rightarrow \vec{AB}, \vec{AC}, \vec{AD}$ lin. ind.

$\Rightarrow K : (A, \vec{AB}, \vec{AC}, \vec{AD})$

$\downarrow$
 $3! = 6$ ways.

• Choose origin in 4 ways (A/B/C/D) $\Rightarrow 4 \cdot 3! = 24$ Cartesian frames.

• Order in $3! = 6$ ways

Problem 3 $K = (O, B)$, $K' = (O', B')$ two frames in A^n .

$$\begin{aligned} (4) P \in A^n, [P]_{K'} &= M_{B'B} \cdot ([P]_K - [O']_K) \\ &= M_{B'B} \cdot ([P]_K - [O']_K) \\ &= M_{B'B} \cdot [P]_K + [O]_{K'} \end{aligned}$$

$$\bullet [P]_{K'} = [O'P]_{B'}$$

$$[P]_K = [OP]_B \quad (1)$$

$$[O']_K = [OO']_B \quad (2)$$

$$\bullet \vec{O'P} = \vec{OP} - \vec{OO'} \quad | \quad B$$

$$[\vec{O'P}]_B = [\vec{OP}]_B - [\vec{OO'}]_B \xrightarrow{(1)} [P]_K - [O']_K \quad (3)$$

$$\bullet [\vec{O'P}]_{B'} = M_{B'B} \cdot [\vec{O'P}]_B$$

$$\xrightarrow{(3)} M_{B'B} \cdot ([P]_K - [O']_K)$$

$$= M_{B'B} \cdot ([P]_K - [O']_K)$$

$$= M_{B'B} \cdot [P]_K - M_{B'B} \cdot [O']_K$$

$$\xrightarrow{(2)} M_{B'B} \cdot [P]_K - M_{B'B} \cdot [OO']_B$$

$$= M_{B'B} \cdot [P]_K - [OO']_{B'}$$

$$= M_{B'B} \cdot [P]_K + [O'O]_{B'}$$

$$= M_{B'B} \cdot [P]_K + [O]_{K'}$$

Problem 4 $l_1: \begin{cases} a_1x + a_2y + a_3z + a_4 = 0 \\ b_1x + b_2y + b_3z + b_4 = 0 \end{cases}, l_2: \begin{cases} c_1x + c_2y + c_3z + c_4 = 0 \\ d_1x + d_2y + d_3z + d_4 = 0 \end{cases}$

l_1, l_2 coplanar $\Leftrightarrow \begin{vmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ c_1 & c_2 & c_3 & c_4 \\ d_1 & d_2 & d_3 & d_4 \end{vmatrix} = 0$; Denote $\det(\tilde{M}) = 0$.

$$\pi_1: a_1x + a_2y + a_3z + a_4 = 0 \quad \pi_2: b_1x + b_2y + b_3z + b_4 = 0 \quad \Rightarrow l_1 = \pi_1 \cap \pi_2.$$

$$\pi_3: c_1x + c_2y + c_3z + c_4 = 0 \quad \pi_4: d_1x + d_2y + d_3z + d_4 = 0 \quad \Rightarrow l_2 = \pi_3 \cap \pi_4.$$

$$\pi_3: c_1x + c_2y + c_3z + c_4 = 0 \quad \pi_4: d_1x + d_2y + d_3z + d_4 = 0$$

$$\pi_4: d_1x + d_2y + d_3z + d_4 = 0$$

$\Rightarrow$ " l_1, l_2 are coplanar. (hyp.)

? $\det(\tilde{M}) = 0$ (cl.)

• case 1: $l_1 \cap l_2 \neq \emptyset$

$\Rightarrow \begin{cases} l_1 \\ l_2 \end{cases}$ has a unique solution.

$\Rightarrow \left. \begin{aligned} \text{rank } M &= \text{rank } \tilde{M} \\ \text{rank} &\leq 3 \end{aligned} \right\} \Rightarrow$

$\Rightarrow \text{rank } \tilde{M} \leq 3 \Rightarrow \det(\tilde{M}) = 0.$
 $\neq 4$

• case 2: $l_1 \parallel l_2$

(def.) $D(l_1) = D(l_2) \Rightarrow \langle v_1 \rangle = \langle v_2 \rangle$

$\Rightarrow D(l_1), D(l_2)$ are solutions of the homogeneous systems.

$\Rightarrow \tilde{M} \xrightarrow{\text{echelon form}} \begin{pmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \text{rank } \tilde{M} < 4$
 $\Rightarrow \det \tilde{M} = 0.$

" $\Leftarrow$ " $\det(\tilde{M}) = 0$ (hyp.)

l_1, l_2 coplanar (cl.)

$\det(\tilde{M}) = 0 \Rightarrow \text{rank}(\tilde{M}) \leq 3.$

l_1, l_2 are lines $\Rightarrow \text{rank } M \geq 2.$

• case 1: $\text{rank}(M) = \text{rank}(\tilde{M}) = 3$ (max) $\Rightarrow$ in 1 point.

$\Rightarrow \begin{cases} l_1 \\ l_2 \end{cases}$ has a unique solution.

$\Rightarrow \exists P \in l_1 \cap l_2 \Rightarrow l_1, l_2$ are coplanar. (they intersect)

• case 2: $\text{rank}(M) = 2$; $\text{rank}(\tilde{M}) = 3.$

$\Rightarrow$ homogeneous eq. for $D(l_2)$ are lin. dep. on eq. of $D(l_1)$

$\Rightarrow D(l_1) = D(l_2)$ direction spaces.

$\Rightarrow l_1 \parallel l_2 \Rightarrow l_1, l_2$ are coplanar.

$\Rightarrow \tilde{M} \sim \begin{pmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ 0 & 0 & 0 & * \\ 0 & 0 & 0 & 0 \end{pmatrix}$

✓ Problem 5 Relative pos. of planes in terms of the rank of the matrix

$$(S) \begin{cases} \pi_1: a_1x + a_2y + a_3z + a_4 = 0 \\ \pi_2: b_1x + b_2y + b_3z + b_4 = 0 \\ \pi_3: c_1x + c_2y + c_3z + c_4 = 0 \end{cases}$$

$$\text{Let } M = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}, \quad \tilde{M} = \begin{bmatrix} a_1 & a_2 & a_3 & -a_4 \\ b_1 & b_2 & b_3 & -b_4 \\ c_1 & c_2 & c_3 & -c_4 \end{bmatrix}.$$

The set of common points: $\pi_1 \cap \pi_2 \cap \pi_3$.

By
Theorem
?

If (S) compatible $\Rightarrow$ the solution set is an affine subspace with dimension $3 - \text{rank}(M)$.

$$(S) \text{ comp. } (\Rightarrow) \text{rank}(M) = \text{rank}(\tilde{M}).$$

If (S) incomp $\Rightarrow$ no common point. (solution set: $\emptyset$).

• case 1: $\text{rank } M < \text{rank } \tilde{M}$

$$\Rightarrow \text{no solution} \Rightarrow \pi_1 \cap \pi_2 \cap \pi_3 \neq \emptyset.$$

• case 2: $\text{rank } M = \text{rank } \tilde{M} = 3$.

$$\Rightarrow (S) \text{ compatible} \Rightarrow \text{solution has dimension } 3 - 3 = 0.$$

$$\Rightarrow 0\text{-dimensional affine space} \Rightarrow \text{point}.$$

$$\Rightarrow \pi_1 \cap \pi_2 \cap \pi_3 = \{P\}. \quad \underline{P(x, y, z)?} \quad \checkmark$$

• case 3: $\text{rank } M = \text{rank } \tilde{M} = 2$.

$$\Rightarrow (S) \text{ compatible} \Rightarrow \dim 3 - 2 = 1.$$

$$\Rightarrow 1\text{-dimensional affine space} \Rightarrow \text{line}.$$

$$\Rightarrow \pi_1 \cap \pi_2 \cap \pi_3 = \{l\},$$

$$l: a_1x + a_2y + a_3z + a_4 = 0 \quad \checkmark$$

• case 4: $\text{rank } M = \text{rank } \tilde{M} = 1$.

$$\Rightarrow (S) \text{ compatible} \Rightarrow \dim 3 - 1 = 2.$$

$$\Rightarrow 2\text{-dimensional affine space} \Rightarrow \text{plane}.$$

$$\Rightarrow \pi_1 \cap \pi_2 \cap \pi_3 = \{\pi_4\}$$

$$? \quad \underline{\pi_4: b_1x + b_2y + b_3z + b_4 = 0} \quad \checkmark$$

Steu 6 ✓ Cartesian equation of $\mathbb{A}^3$ $\rightarrow$ parametric equations.

Freudenberg

plane in Cartesian form: $\bar{u}: ax + by + cz + d = 0$

plane in parametric form: $(x, y, z) = P_0 + s \cdot \vec{u} + t \cdot \vec{v}$

$$\vec{u} = (u_1, u_2, u_3), \vec{v} = (v_1, v_2, v_3).$$

• C $\rightarrow$ P:

• find P_0 on the plane: $y=0, z=0$

$$\Rightarrow ax + d = 0 \Rightarrow x = -\frac{d}{a} \quad (a \neq 0)$$

$$\Rightarrow P_0(x_0, y_0, z_0) \Rightarrow P_0\left(-\frac{d}{a}, 0, 0\right)$$

• find direction vectors: $ax + by + cz = 0$.

$$y=1, z=0 \Rightarrow ax + b = 0 \Rightarrow x = -\frac{b}{a} \Rightarrow \vec{u}\left(-\frac{b}{a}, 1, 0\right).$$

$$y=0, z=1 \Rightarrow ax + c = 0 \Rightarrow x = -\frac{c}{a} \Rightarrow \vec{v}\left(-\frac{c}{a}, 0, 1\right).$$

• check lin. indep:

$$\alpha\left(-\frac{b}{a}, 1, 0\right) + \beta\left(-\frac{c}{a}, 0, 1\right) = 0$$

$$\Rightarrow \left(-\alpha \cdot \frac{b}{a} - \beta \cdot \frac{c}{a}, \alpha, \beta\right) = 0$$

$$\Rightarrow \begin{cases} -\alpha \cdot \frac{b}{a} = 0 \\ -\beta \cdot \frac{c}{a} = 0 \end{cases} \quad \begin{matrix} -\alpha \cdot \frac{b}{a} - \beta \cdot \frac{c}{a} = 0 \\ \alpha = 0 \\ \beta = 0 \end{matrix} \quad \begin{matrix} \vec{u}, \vec{v} \\ \Rightarrow \text{lin. indep.} \end{matrix}$$

$$\Rightarrow (x, y, z) = \left(-\frac{d}{a}, 0, 0\right) + s \cdot \left(-\frac{b}{a}, 1, 0\right) + t \cdot \left(-\frac{c}{a}, 0, 1\right).$$

• P $\rightarrow$ C:

• find normal vector: $\vec{n} = \vec{u} \times \vec{v} \Rightarrow \vec{n} = (a, b, c)$

• use point-normal equation: $a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$.

$$\Rightarrow ax + by + cz - (ax_0 + by_0 + cz_0) = 0.$$

$$\Rightarrow - (ax_0 + by_0 + cz_0) = d \quad \Rightarrow ax + by + cz + d = 0.$$

• expand: $ax + by + cz + d = 0$.

$$\Rightarrow ax + by + cz + d = 0.$$

? **Problem 7** $K = (0, B)$ in A^m .

$$(s) \begin{cases} a_{11}x_1 + \dots + a_{1m}x_m = b_1 \\ \vdots \\ a_{t1}x_1 + \dots + a_{tm}x_m = b_t \end{cases}$$

? $S \rightarrow$ affine space of dim $d =$
 ? $D(s) \rightarrow$ vector subspace of V^m

$$D(s): \begin{cases} a_{11}x_1 + \dots + a_{1m}x_m = 0 \\ \vdots \\ a_{t1}x_1 + \dots + a_{tm}x_m = 0 \end{cases}$$

Theorem 2.6.

$\Rightarrow$ " S set of solutions to (s) ; $S = \{x_1, \dots, x_m\}$. (hyp.)

S -affine subspace of A^m (cl.)

• We show $S = U + P$.

$P \rightarrow$ solution from S ; $P(p_1, \dots, p_m)$

$U \rightarrow$ vector subspace.

• We show $S = T$.

• We show $S \subseteq T$.

$T \rightarrow$ sol. from S

$Q \rightarrow$ sol from S ; $Q(q_1, \dots, q_m) \in S$.

$$\Rightarrow \boxed{Q \in T \Leftrightarrow \vec{PQ} \in U}$$

$$\begin{aligned} & a_{j1}(q_1 - p_1) + \dots + a_{jm}(q_m - p_m) = \\ & = \underbrace{a_{j1}q_1 + \dots + a_{jm}q_m}_{b_j} - \underbrace{(a_{j1}p_1 + \dots + a_{jm}p_m)}_{b_j} = 0. \end{aligned}$$

$$\Rightarrow \vec{PQ} \in U.$$

$$\Rightarrow S \subseteq T. \Rightarrow$$

$$T \subseteq S$$

$$\Rightarrow \begin{cases} S \subseteq T \\ T \subseteq S \end{cases} \Rightarrow S = T.$$

$$d. = \dim S = \dim \ker$$

$$\dim = \dim \text{Imf.}$$

$$d - s$$

$$\Rightarrow S = U + P.$$

$$\text{dimension} = d.$$

Problem 8 $\triangle ABC$; $|AB|=c$, $|BC|=a$, $|CA|=b$.

$$\therefore \langle \vec{AB}, \vec{AC} \rangle = \frac{1}{2}(b^2 + c^2 - a^2)$$

? cos law.

$$\vec{CA} + \vec{AB} = \vec{CB}$$

$$\Rightarrow \vec{BC} = \vec{AC} - \vec{AB}$$

$$\begin{aligned} |BC|=a \quad |^2 \Rightarrow |BC|^2 = a^2 &= \langle \vec{BC}, \vec{BC} \rangle = \langle \vec{AC} - \vec{AB}, \vec{AC} - \vec{AB} \rangle = \\ &= \langle \vec{AC}, \vec{AC} \rangle - \langle \vec{AC}, \vec{AB} \rangle - \langle \vec{AB}, \vec{AC} \rangle + \langle \vec{AB}, \vec{AB} \rangle = \\ &= |\vec{AC}|^2 + |\vec{AB}|^2 - 2 \cdot \langle \vec{AB}, \vec{AC} \rangle \end{aligned}$$

$$\Rightarrow a^2 = b^2 + c^2 - 2 \cdot \langle \vec{AB}, \vec{AC} \rangle$$

$$\Rightarrow \langle \vec{AB}, \vec{AC} \rangle = \frac{1}{2}(b^2 + c^2 - a^2)$$

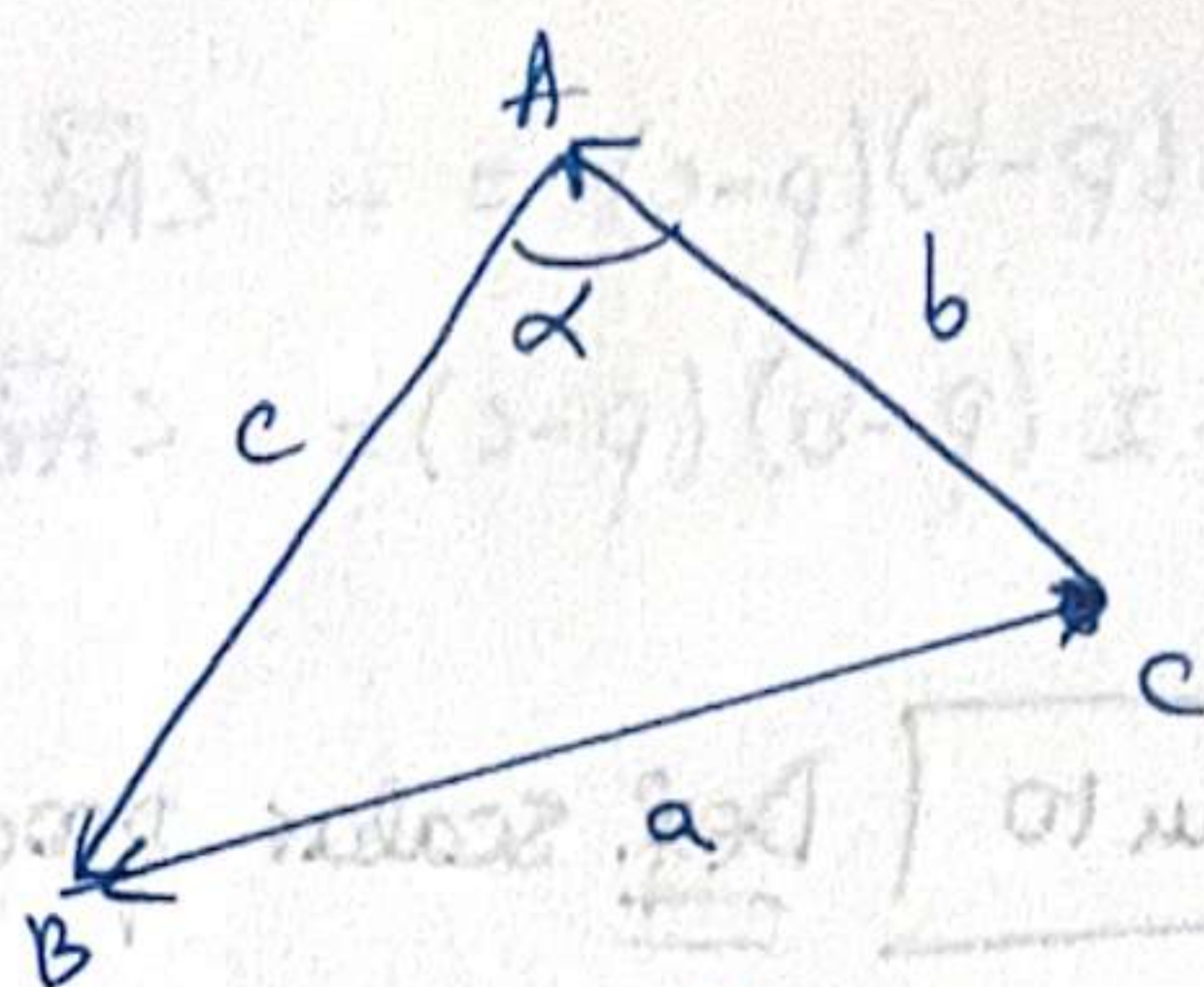
$\alpha = \angle BAC$.

$$\langle \vec{AB}, \vec{AC} \rangle = |AB| \cdot |AC| \cdot \cos \alpha = \frac{1}{2}(b^2 + c^2 - a^2)$$

$$\Leftrightarrow c \cdot b \cdot \cos \alpha = \frac{1}{2}(b^2 + c^2 - a^2) \quad | \cdot 2$$

$$\Leftrightarrow 2bc \cdot \cos \alpha = b^2 + c^2 - a^2$$

$$\Leftrightarrow \underline{a^2 = b^2 + c^2 - 2bc \cdot \cos \alpha}$$



Problem 9 $\triangle ABC$, $|AB|=c$, $|BC|=a$, $|CA|=b$, $p = \frac{1}{2}(a+b+c)$.

$$\langle \vec{AB}, \vec{AC} \rangle + bc = 2p(p-a)$$

$$\langle \vec{AB}, \vec{AC} \rangle - bc = -2(p-b)(p-c)$$

$$p = \frac{1}{2}(a+b+c)$$

$$p-a = \frac{1}{2}a + \frac{1}{2}b + \frac{1}{2}c - a = \frac{1}{2}(-a+b+c) \quad | \cdot 4p$$

$$\Rightarrow 4p(p-a) = p(-a+b+c) = (a+b+c)(-a+b+c) = (b+c)^2 - a^2$$

$$= b^2 + c^2 + 2bc - a^2$$

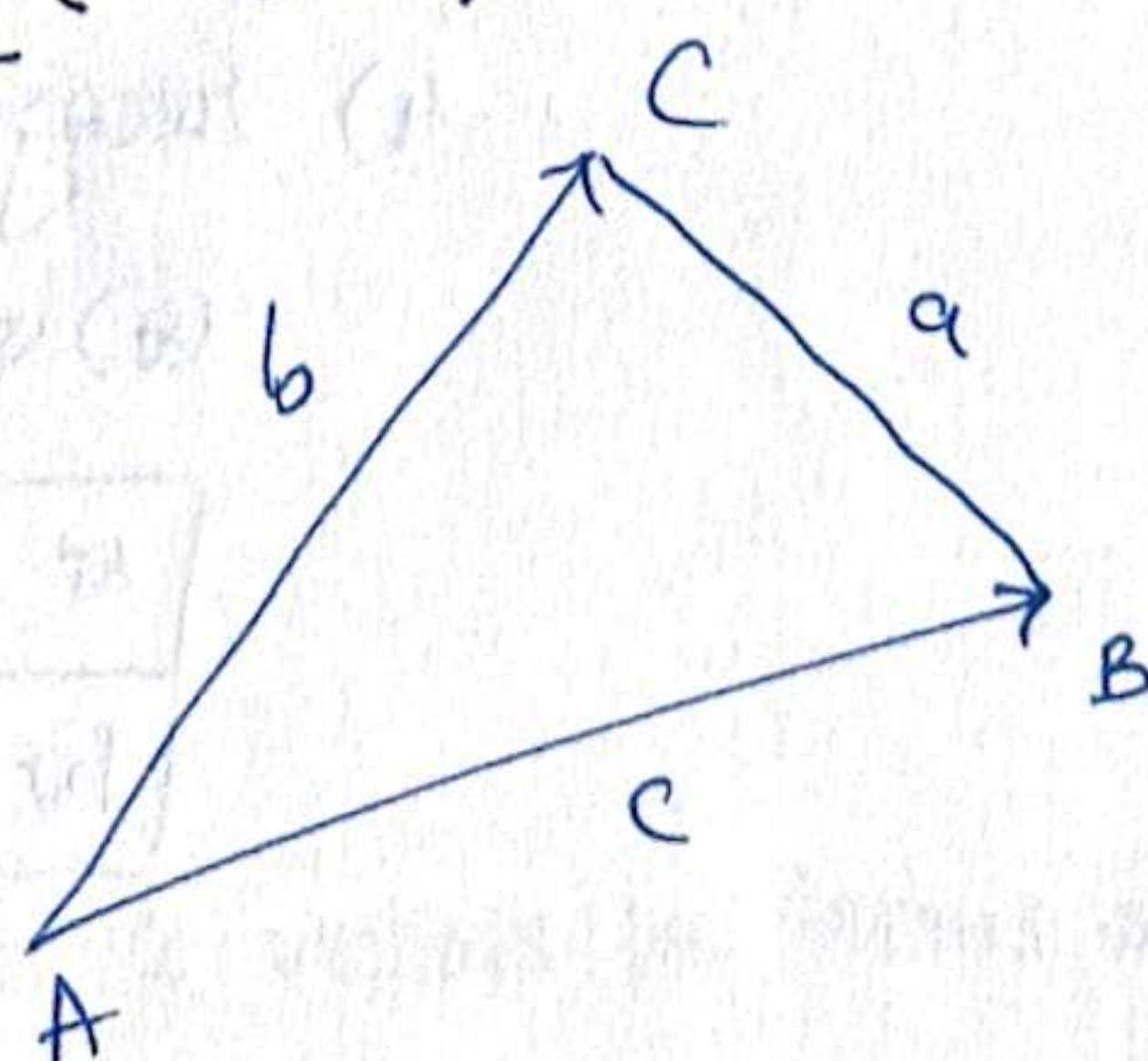
$$= -a^2 + b^2 + c^2 + 2bc$$

$$= 2 \cdot \langle \vec{AC}, \vec{AB} \rangle + 2bc$$

$$4(p-b)(p-c) = (a-b+c)(a+b-c) = a^2 - (b-c)^2 = \underline{a^2 - b^2 - c^2 + 2bc}$$

$$= -2 \langle \vec{AC}, \vec{AB} \rangle + 2bc \quad | : 2$$

$$\boxed{\langle \vec{AB}, \vec{AC} \rangle = \frac{1}{2}(b^2 + c^2 - a^2)}$$



$$\Rightarrow 2(p-b)(p-c) = - \langle \vec{AC}, \vec{AB} \rangle + bc \cdot 1 \cdot (-1) = |b||c| \cdot (-1) = -|b||c|$$

$$\Rightarrow -2(p-b)(p-c) = \langle \vec{AC}, \vec{AB} \rangle - bc \cdot \frac{(-5+5+5)}{5} = \langle \vec{AC}, \vec{AB} \rangle - bc$$

Problem 10 | Def. scalar prod. Prop. scalar prod. Deduce prop. from def.

$$\text{Def: } \langle a, b \rangle = \begin{cases} 0, & \text{if } a=0 \text{ or } b=0 \\ |a| \cdot |b| \cdot \cos \angle(a, b), & \text{if } a \neq 0, b \neq 0. \end{cases}$$

$$\langle -, - \rangle : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}.$$

map

Properties: 1) bilinear:

$$(\forall) a, b \in \mathbb{R}, (\forall) u, w, v \in \mathbb{N}^2:$$

$$\boxed{\begin{aligned} \langle a\underline{v} + b\underline{w}, \underline{u} \rangle &= a \langle \underline{v}, \underline{u} \rangle + b \langle \underline{w}, \underline{u} \rangle \\ \langle \underline{v}, a\underline{w} + b\underline{u} \rangle &= a \langle \underline{v}, \underline{w} \rangle + b \langle \underline{v}, \underline{u} \rangle \end{aligned}}$$

2) symmetric:

$$(\forall) \underline{v}, \underline{w} \in \mathbb{N}^2:$$

$$\boxed{\langle \underline{v}, \underline{w} \rangle = \langle \underline{w}, \underline{v} \rangle}$$

3) positive-definite:

$$(\forall) \underline{v} \in \mathbb{N}^2:$$

$$\boxed{\underline{v} \neq 0 \Rightarrow \langle \underline{v}, \underline{v} \rangle > 0.}$$

4) recognizes right angles and unit lengths.

$$(\forall) \underline{v}, \underline{w} \in \mathbb{N}^2:$$

$$\boxed{\begin{aligned} \underline{v} \perp \underline{w} &(\Leftrightarrow) \langle \underline{v}, \underline{w} \rangle = 0 \\ |\underline{v}| = 1 &(\Leftrightarrow) \langle \underline{v}, \underline{v} \rangle = 1 \end{aligned}}$$

Prove bilinearity: by symmetry, $\langle a, b \rangle = |a| \cdot |b| \cdot \cos \angle(a, b)$.

$$= |b| \cdot |a| \cdot \frac{\text{pr}_{\underline{b}}^{\perp}(\underline{a})}{|a|}$$

$$= |b| \cdot \underbrace{\text{pr}_{\underline{b}}^{\perp}(\underline{a})}_{\text{ct. linear fct. in } b}.$$

Prove symmetry:

ove pos-def:

Right-angles, unit lengths:

Remarks: $\text{Pr}_a^\perp b = \frac{\langle a, b \rangle}{\langle a, a \rangle} \cdot a$

$$\text{Pr}_b^\perp a = \frac{\langle a, b \rangle}{\langle b, b \rangle} \cdot b$$

$$\text{Pr}_a^\perp(b) = \underbrace{\text{Pr}_a^\perp(b)} \cdot \frac{a}{|a|}$$

$$= |b| \cdot \cos \angle(a, b) \cdot \frac{a}{|a|}$$

$$= \frac{|b| \cdot \cos \angle(a, b)}{|a|} \cdot \frac{|a| \cdot a}{|a|}$$

$$= \underbrace{\frac{\text{Pr}_a^\perp b}{|b|}}_{\cos} \cdot |b| \cdot \frac{a}{|a|}$$

✓ **Problem 12** Scalar prod. in dim n . Scalar prod. of 2 vectors in terms of their coordinates in an orthonormal basis.

$$\langle a, b \rangle = \begin{cases} 0, & \text{if } a=0 \text{ or } b=0 \\ |a| \cdot |b| \cdot \cos \angle(a, b), & \text{otherwise.} \end{cases}$$

orthonormal basis: $a = (a_1, \dots, a_m)$, $b = (b_1, \dots, b_m)$:

$$\langle a, b \rangle = \sum_{i=1}^m a_i \cdot b_i.$$

Problem 12 / $B = (e_1, e_2, \dots, e_m)$ of $\mathbb{R}^m$. Construct B' :

$$e'_1 = e_1$$

$$e'_2 = e_2 - \frac{\langle e'_1, e_2 \rangle}{\langle e'_1, e'_1 \rangle} \cdot e'_1 = e_2 - \text{Pr}_{e'_1}^\perp(e_2)$$

$$e'_3 = e_3 - \frac{\langle e'_1, e_3 \rangle}{\langle e'_1, e'_1 \rangle} \cdot e'_1 - \frac{\langle e'_2, e_3 \rangle}{\langle e'_2, e'_2 \rangle} \cdot e'_2 = e_3 - \text{Pr}_{e'_1}^\perp(e_3) - \text{Pr}_{e'_2}^\perp(e_3)$$

$$\Rightarrow B' = \left\{ \frac{e'_1}{\|e'_1\|}, \dots, \frac{e'_m}{\|e'_m\|} \right\} \text{ orthonormal basis?}$$

Proof of Gram-Schmidt

• $\forall k \Rightarrow W_k$ generated by $B_k = (e_1, \dots, e_k)$.

$\Rightarrow B_{k'} = (e'_1, \dots, e'_k)$ orthonormal basis for W_k .

• $k=1 \Rightarrow$ nothing to prove.

• $k>1 \Rightarrow$ consider:

$$e'_{k+1} = e_{k+1} - \underbrace{\sum_{j=1}^k \text{Pr}_{e'_j}(e_{k+1})}_{\substack{\neq e_{k+1} \Rightarrow e_{k+1} \text{ lin. ind. of } W_k}}$$

• linear combination of the first k vectors:

$$\frac{\langle e'_j, e_{k+1} \rangle}{\langle e'_j, e'_j \rangle} \cdot e'_j$$

$(e'_1, \dots, e'_k, e_{k+1})$

lin. ind + generate the space $\Rightarrow$ basis of W_k .

$$\langle e'_{k+1}, e'_i \rangle = \langle e_{k+1} - \sum_{j=1}^k \frac{\langle e'_j, e_{k+1} \rangle}{\langle e'_j, e'_j \rangle} \cdot e'_j, e'_i \rangle$$

$$= \langle e_{k+1}, e'_i \rangle - \sum_{j=1}^k \frac{\langle e'_j, e_{k+1} \rangle}{\langle e'_j, e'_j \rangle} \cdot \underbrace{\langle e'_j, e'_i \rangle}_{\substack{1 \text{ if } i=j \\ 0 \text{ otherwise.}}}$$

$$= \langle e_{k+1}, e'_i \rangle - \frac{\langle e'_i, e_{k+1} \rangle}{\langle e'_i, e'_i \rangle} \cdot \langle e'_i, e'_i \rangle$$

$$= 0$$

Problem 13

Normal vector to a hyperplane in dimension n .

How normal vectors relate to the Cartesian eq. of a hyperplane.

- A vector $m \neq 0$ normal to $H \subset V^m \Leftrightarrow \forall u \in \vec{H} : \langle m, u \rangle = 0$
(m is orthogonal to all dir. vectors of H)

- Cartesian eq: $H = \{x \in V^m \mid \langle m, \vec{Ax} \rangle = 0\}$

- Coordinate form: $m = (a_1, \dots, a_m)$

$$x = (x_1, \dots, x_m)$$

$$A = (x_1^0, \dots, x_m^0)$$

$$\sum_{i=1}^m a_i (x_i - x_i^0) = 0 \Leftrightarrow \sum_{i=1}^m a_i x_i + d = 0.$$

$$x \in H \Leftrightarrow \vec{Ax} \in \vec{H}$$

$$m \text{ normal} \Rightarrow \langle m, \vec{Ax} \rangle = 0$$

scalar pr.
orth. basis

$$\sum a_i (x_i - x_i^0) = 0 \Rightarrow \text{eq. } H.$$

Problem 14

Normal vector = vector orthogonal to the affine space.

Hyperplane: $\mathcal{H} : a_1 x_1 + \dots + a_m x_m = b$ (try)

K orthonormal frame $\Rightarrow \mathcal{B}(e_1, \dots, e_m)$ orthonormal.

Fix $Q(q_1, \dots, q_m)$ in $\mathcal{H}$ (try)

$$\Rightarrow b = a_1 q_1 + \dots + a_m q_m \quad (1)$$

$P(p_1, \dots, p_m)$ in $\mathcal{H}$

$$\Rightarrow b = a_1 p_1 + \dots + a_m p_m \quad (2)$$

$$(2) - (1) \Rightarrow P - Q = \vec{QP} = a_1 (p_1 - q_1) + \dots + a_m (p_m - q_m) = 0.$$

$$m = (a_1, \dots, a_m) \Rightarrow P \in \mathcal{H} \Leftrightarrow \langle m, \vec{QP} \rangle = 0$$

$$\Leftrightarrow m \perp \vec{QP}.$$

Problem 14

$d(P, \mathcal{H})$ in $\dim m$; $K \rightarrow$ orthonormal frame
 $B \rightarrow$ fixed orthonormal basis

$$d(P, \mathcal{H}) = \frac{|\sum_{i=1}^m a_i x_i^0 + d|}{\sqrt{\sum_{i=1}^m a_i^2}}$$

$A \in \mathcal{H} \Rightarrow \mathcal{H} = \{x \mid \langle m, \vec{Ax} \rangle = 0\}$

distance = length of orthogonal projection: $d(P, \mathcal{H}) = \frac{|\langle m, \vec{AP} \rangle|}{|m|}$

??! $\langle m, \vec{AP} \rangle = \sum a_i (x_i^0 - x_i^A)$

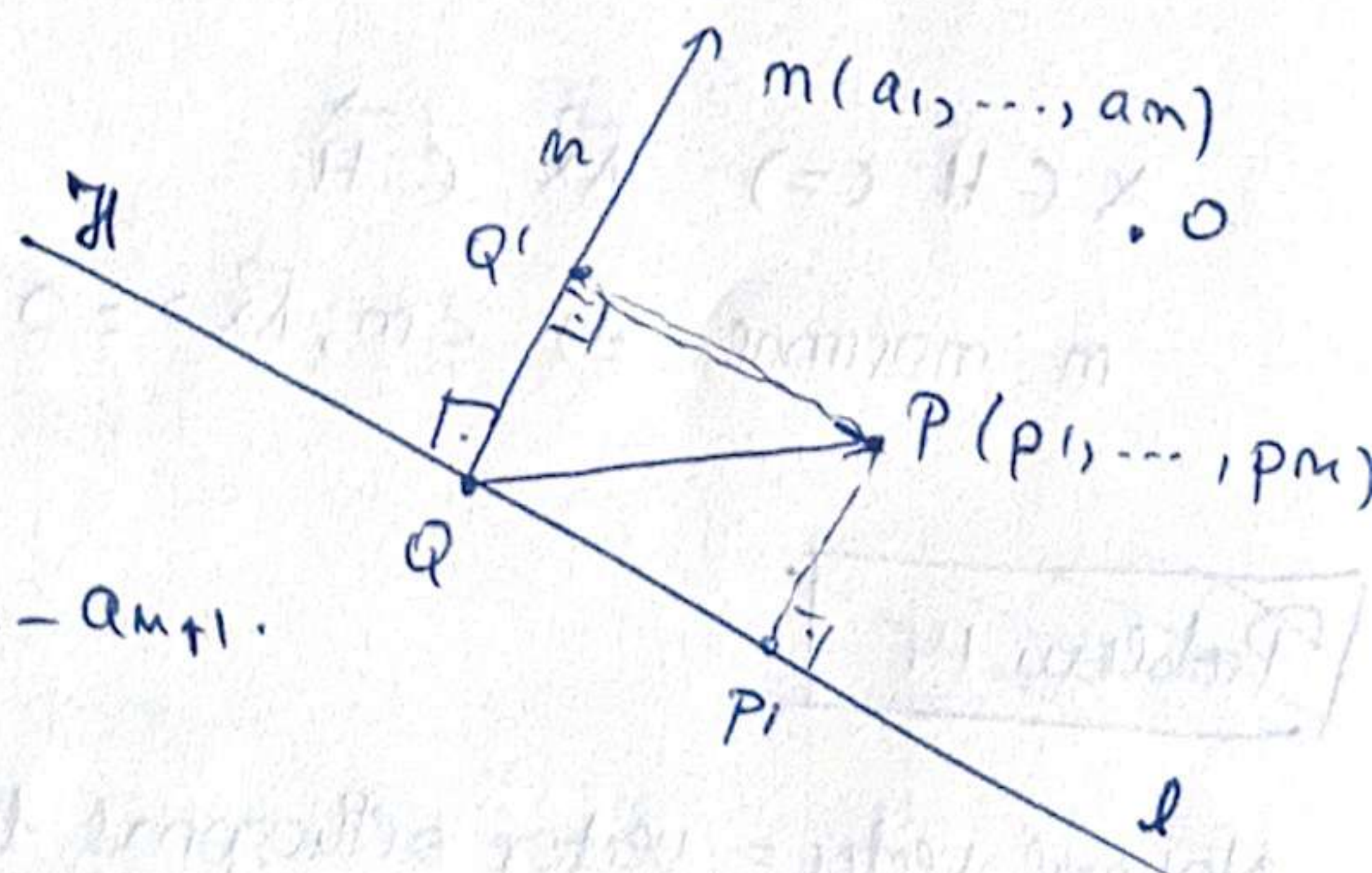
$\mathcal{H}: \sum a_i x_i^A + d = 0 \Rightarrow d(P, \mathcal{H}) = \frac{|\sum a_i x_i^0 + d|}{\sqrt{\sum a_i^2}}$

$$d(P, \mathcal{H}) = \frac{|a_1 p_1 + \dots + a_m p_m + a_{m+1}|}{\sqrt{a_1^2 + \dots + a_m^2}}$$

$\mathcal{H}: a_1 x_1 + \dots + a_m x_m + a_{m+1} = 0.$

$Q(g_1, \dots, g_m) \in \mathcal{H} \Rightarrow a_1 g_1 + \dots + a_m g_m = -a_{m+1}.$

$m(a_1, \dots, a_m)$ normal vector.



$$\begin{aligned} d(P, \mathcal{H}) &= |PQ'| = |Q'Q| = |\text{Proj}_m(\vec{QP})| = \left| \frac{\langle m, \vec{QP} \rangle}{\langle m, m \rangle} \cdot m \right| = \\ &= \left| \frac{\langle m, \vec{QP} \rangle}{\langle m, m \rangle} \cdot |m| \right| = \left| \frac{\langle m, \vec{QP} \rangle}{|m|^2} \cdot |m| \right| = \left| \frac{\langle m, \vec{QP} \rangle}{|m|} \right| = \end{aligned}$$

$\vec{QP} = \vec{QP}' + \vec{P'Q}$

$\vec{QP}' + \vec{P'Q} = \vec{QP}$

$\vec{OP} - \vec{OQ} = \vec{QP}$

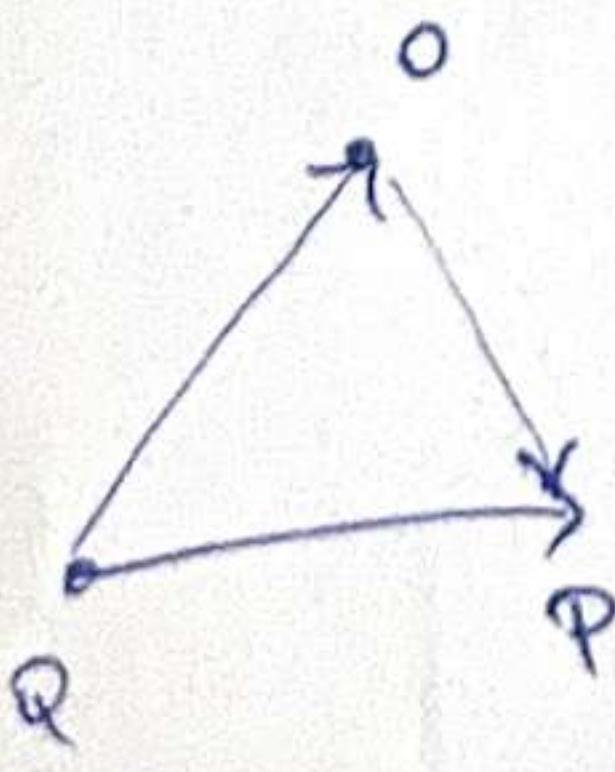
$\vec{QO} + \vec{OP} = \vec{QP}$
 $-\vec{OQ} + \vec{OP}$

$$= \left| \frac{\langle m, \vec{OP} - \vec{OQ} \rangle}{|m|} \right| = \left| \frac{\langle m, \vec{OP} \rangle - \langle m, \vec{OQ} \rangle}{|m|} \right|$$

$$= \frac{|a_1 p_1 + \dots + a_m p_m - (a_1 g_1 + \dots + a_m g_m)|}{\sqrt{a_1^2 + \dots + a_m^2}}$$

$$= \frac{|a_1 p_1 + \dots + a_m p_m - (-a_{m+1})|}{\sqrt{a_1^2 + \dots + a_m^2}}$$

$$= \frac{|a_1 p_1 + \dots + a_{m+1}|}{\sqrt{a_1^2 + \dots + a_m^2}}$$



5] Points in $\mathbb{E}^2$ that are equidistant from a given affine subspace.

Sol: Case I: Affine subspace = point: dist between two points

$$\begin{aligned} \mathcal{L} &: \{P \in \mathbb{E}^2 \mid \text{dist}(P, Q) = \alpha\} = \\ &= \{P(x, y) \mid \sqrt{(x-x_Q)^2 + (y-y_Q)^2} = \alpha\} = \text{circle}. \end{aligned}$$

Case II: Affine subspace = line ℓ :

$$\begin{aligned} \mathcal{L} &: \{P \in \mathbb{E}^2 \mid \text{dist}(P, \ell) = \alpha\} \\ &= \{P(x, y) \mid \frac{|ax + by + c|}{\sqrt{a^2 + b^2}} = \alpha\} \end{aligned}$$

dist between a line and a point.

$$= \{P \mid ax + by + c = \pm \alpha \cdot \sqrt{a^2 + b^2}\}$$

$$= \{P \mid ax + by + c + \alpha \cdot \sqrt{a^2 + b^2} = 0\} \cup$$

$$\cup \{P \mid ax + by + c - \alpha \cdot \sqrt{a^2 + b^2} = 0\}$$

= 2 lines. $\Rightarrow \mathcal{U}$.

Problem 16 Def. $\mathcal{J}$ -operator (dim. 2). Deduce formula for $\mathcal{J}(v)$ in terms of coord. v in an orthonormal basis.

• $\mathcal{J}(v) = (a, b)$; $v(x, y) \neq 0 \Rightarrow x^2 + y^2 \neq 0$. (1)

$$\begin{cases} ax + by = 0 \quad (\perp) \Rightarrow b = -\frac{ax}{y}, y \neq 0. \\ x^2 + y^2 = a^2 + b^2 \end{cases}$$

$$\Rightarrow x^2 + y^2 = a^2 + a^2 \cdot \frac{x^2}{y^2} = a^2 \cdot \left(1 + \frac{x^2}{y^2}\right)$$

$$\Rightarrow \underbrace{x^2 + y^2}_{\neq 0} = a^2 - \frac{x^2 + y^2}{b^2} \Rightarrow a^2 = y^2 \Rightarrow a = \pm y.$$

(1)

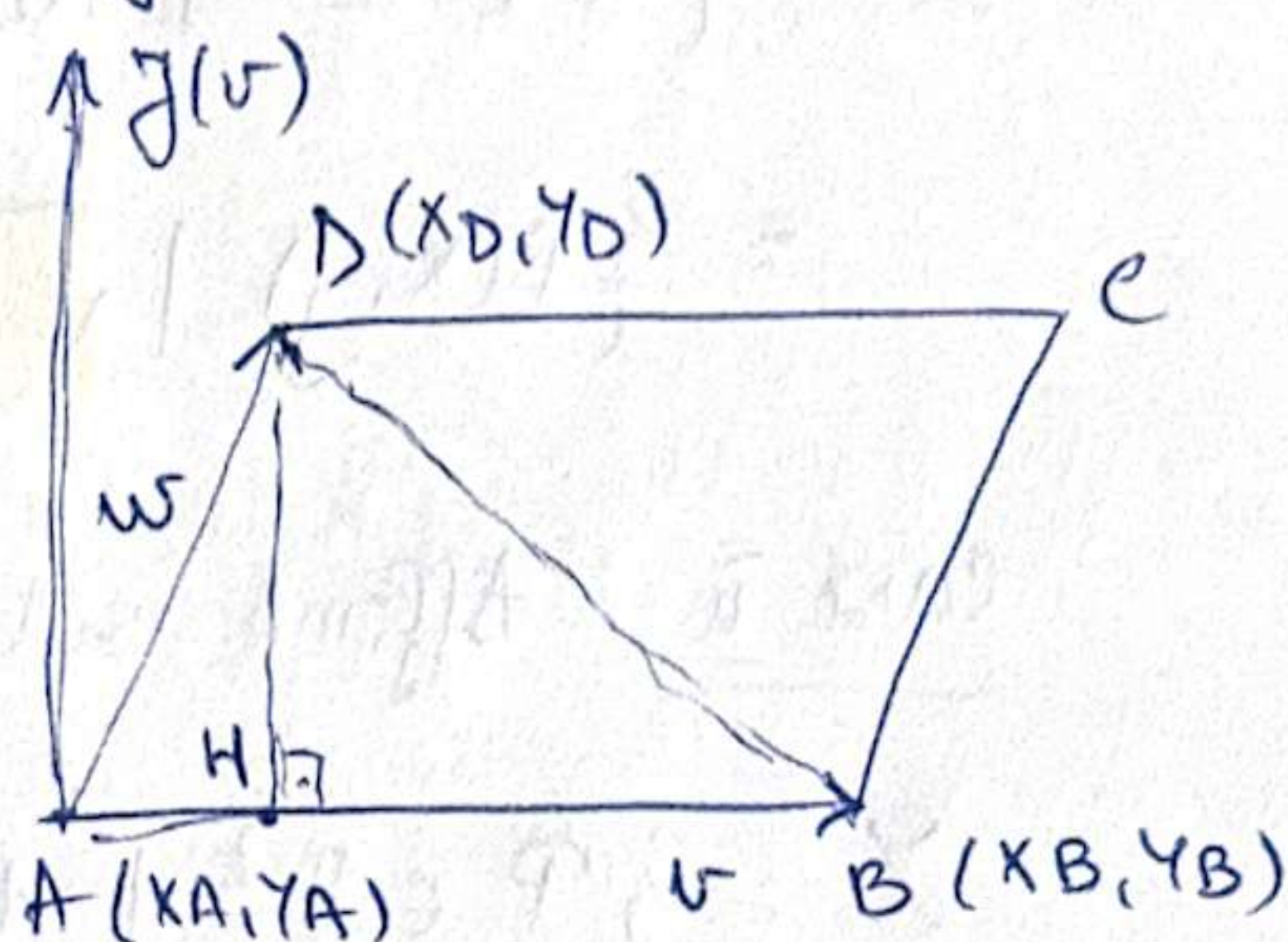
$$\Rightarrow \begin{vmatrix} x & a \\ y & b \end{vmatrix} = \begin{vmatrix} x & \pm y \\ y & \mp y - \frac{x}{y} \end{vmatrix} = \begin{vmatrix} x & \pm y \\ y & \mp x \end{vmatrix} = \mp (x^2 + y^2) > 0.$$

Problem 17. $A(x_A, y_A), B(x_B, y_B), D(x_D, y_D)$ of ABCD parallelogram.

$$A_{ABCD} = \begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_D & y_D & 1 \end{vmatrix} = \begin{vmatrix} v_x & v_y & 1 \\ w_x & w_y & 1 \end{vmatrix}$$

$$v(v_x, v_y) = \vec{AB}$$

$$w(w_x, w_y) = \vec{AD}$$



$$A_{ABCD} = b \cdot h$$

$$= |\vec{AB}| \cdot |\vec{HD}|$$

$$= |v| \cdot |P_{\vec{g}(v)} w| = |v| \cdot \left| \frac{\langle \vec{g}(v), w \rangle}{\langle \vec{g}(v), \vec{g}(v) \rangle} \right| \cdot |\vec{g}(v)|$$

$$? \parallel v$$

$$= |v| \cdot \left| \frac{\langle \vec{g}(v), w \rangle}{|\vec{g}(v)|} \right| = \left| \langle \vec{g}(v), w \rangle \right| = |-v_y \cdot w_x + v_x \cdot w_y|$$

$$= \begin{vmatrix} v_x & v_y \\ w_x & w_y \end{vmatrix} = \begin{vmatrix} x_B - x_A & y_B - y_A \\ x_D - x_A & y_D - y_A \end{vmatrix} = \begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_D & y_D & 1 \end{vmatrix}$$

$$= \begin{vmatrix} x_A & y_A & 1 \\ x_B - x_A & y_B - y_A & 0 \\ x_D - x_A & y_D - y_A & 0 \end{vmatrix} = \begin{vmatrix} x_A & y_A & 1 \\ x_B & y_B & 1 \\ x_D & y_D & 1 \end{vmatrix}$$

Problem 18. $\triangle ABC$. $u = \vec{AB}, v = \vec{BC}, w = \vec{CA}$.

$$u \times v = v \times w = w \times u.$$

sin law

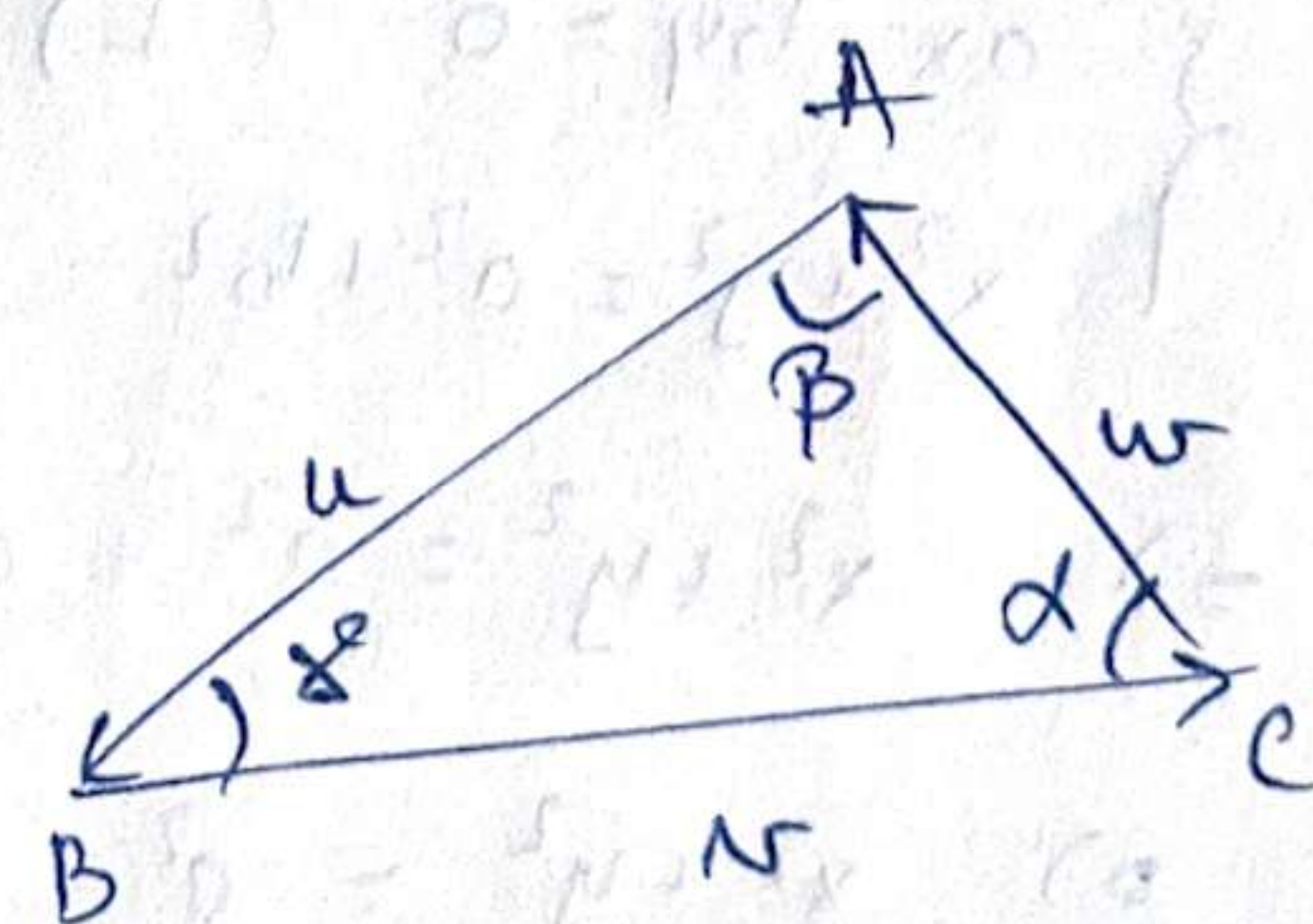
$$u + v + w = 0 \Rightarrow w = -(u + v). \quad (1)$$

$$u \times w = u \times (-(u + v)) = -u \times u - u \times v = u \times v$$

$$w \times u = (-(u + v)) \times u = -u \times u - v \times u = u \times v.$$

$$\Rightarrow u \times v = v \times w = w \times u = A$$

$$\begin{aligned} |u \times v| &= |u| \cdot |v| \cdot \sin \gamma = A \\ |v \times w| &= |v| \cdot |w| \cdot \sin \alpha = A \\ |w \times u| &= |w| \cdot |u| \cdot \sin \beta = A \end{aligned} \Rightarrow |u| \cdot |v| \cdot \sin \gamma = |v| \cdot |w| \cdot \sin \alpha = |w| \cdot |u| \cdot \sin \beta$$



$$|u| \cdot |v| \cdot \sin \gamma = |v| \cdot |w| \cdot \sin \alpha \quad | : |v| \neq 0$$

$$\Rightarrow |u| \cdot \sin \gamma = |w| \cdot \sin \alpha$$

$$\Rightarrow \frac{|u|}{\sin \alpha} = \frac{|w|}{\sin \gamma} \quad (1)$$

$$|v| \cdot |w| \cdot \sin \alpha = |w| \cdot |u| \cdot \sin \beta \quad | : |w| \neq 0$$

$$\Rightarrow \frac{|v|}{\sin \beta} = \frac{|u|}{\sin \alpha} \quad (2)$$

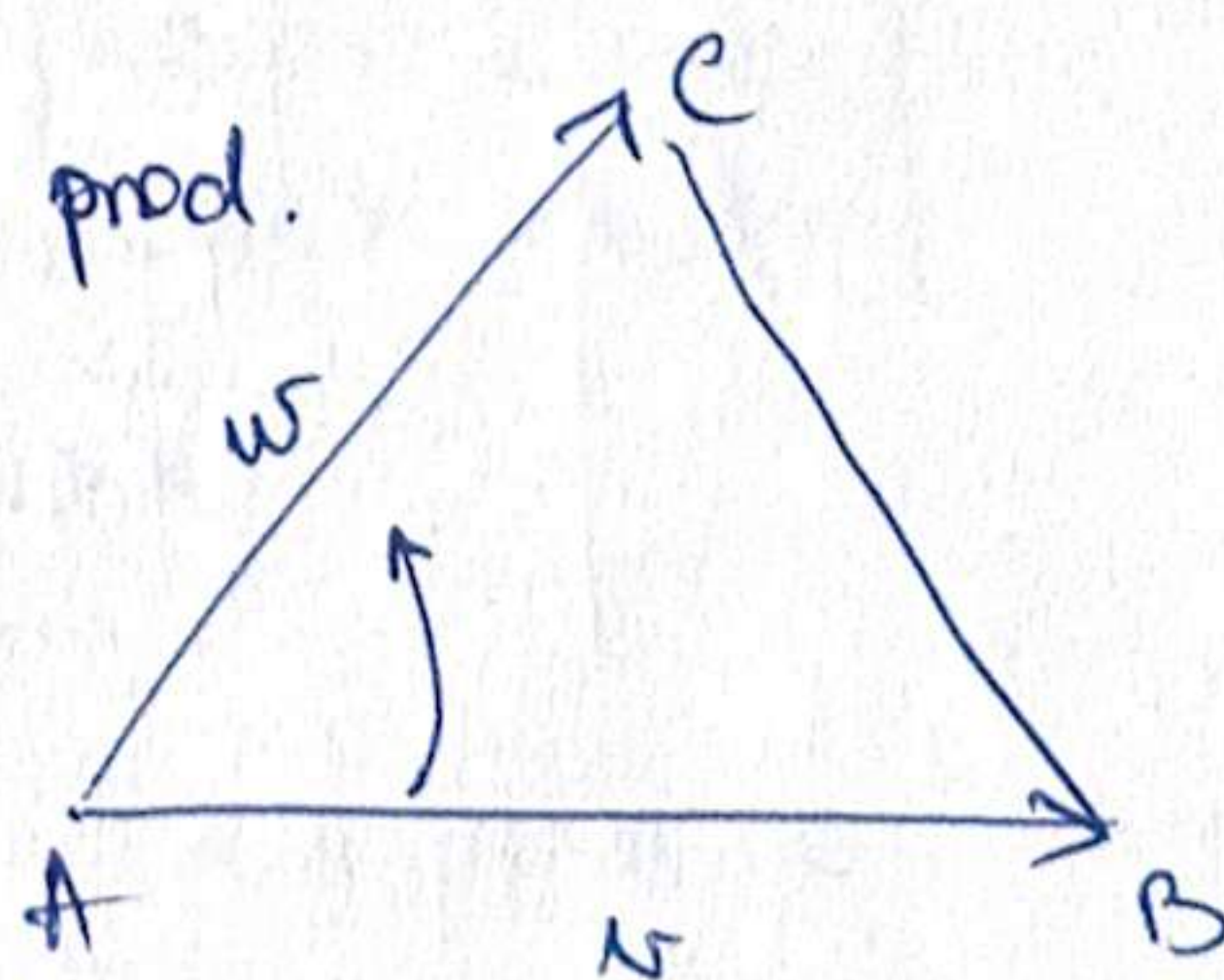
$$(1) = (2) \Rightarrow \frac{|u|}{\sin \alpha} = \frac{|v|}{\sin \beta} = \frac{|w|}{\sin \gamma}$$

Problem 19 ✓ Expression for sine of an oriented angle $\rightarrow$ ^{box} product.
cosine $\rightarrow$ scalar product.

$$(\sin \angle_{\text{or}}(v, w))^2 + (\cos \angle_{\text{or}}(v, w))^2 = 1.$$

$$\sin \angle_{\text{or}}(v, w) = \frac{[v, w]}{|v| \cdot |w|} = \frac{2 \cdot \text{Area}(\triangle ABC)}{|AB| \cdot |AC|} \rightarrow \text{box prod.}$$

$$\cos \angle_{\text{or}}(v, w) = \frac{\langle v, w \rangle}{|v| \cdot |w|} \rightarrow \text{scalar prod.}$$



$$v = \vec{AB}$$

$$w = \vec{AC}$$

$$(\sin \angle_{\text{or}}(v, w))^2 + (\cos \angle_{\text{or}}(v, w))^2 =$$

$$= \frac{[v, w]^2}{|v|^2 \cdot |w|^2} + \frac{\langle v, w \rangle^2}{|v|^2 \cdot |w|^2}$$

$$= \frac{(v_1^2 + v_2^2)(w_1^2 + w_2^2)}{|v|^2 \cdot |w|^2} = \frac{|v|^2 \cdot |w|^2}{|v|^2 \cdot |w|^2} = 1.$$

$$[v, w] = v_1 w_2 - v_2 w_1$$

$$\langle v, w \rangle = v_1 w_1 + v_2 w_2$$

$$[v, w]^2 = (v_1 w_2 - v_2 w_1)^2$$

$$\langle v, w \rangle^2 = (v_1 w_1 + v_2 w_2)^2$$

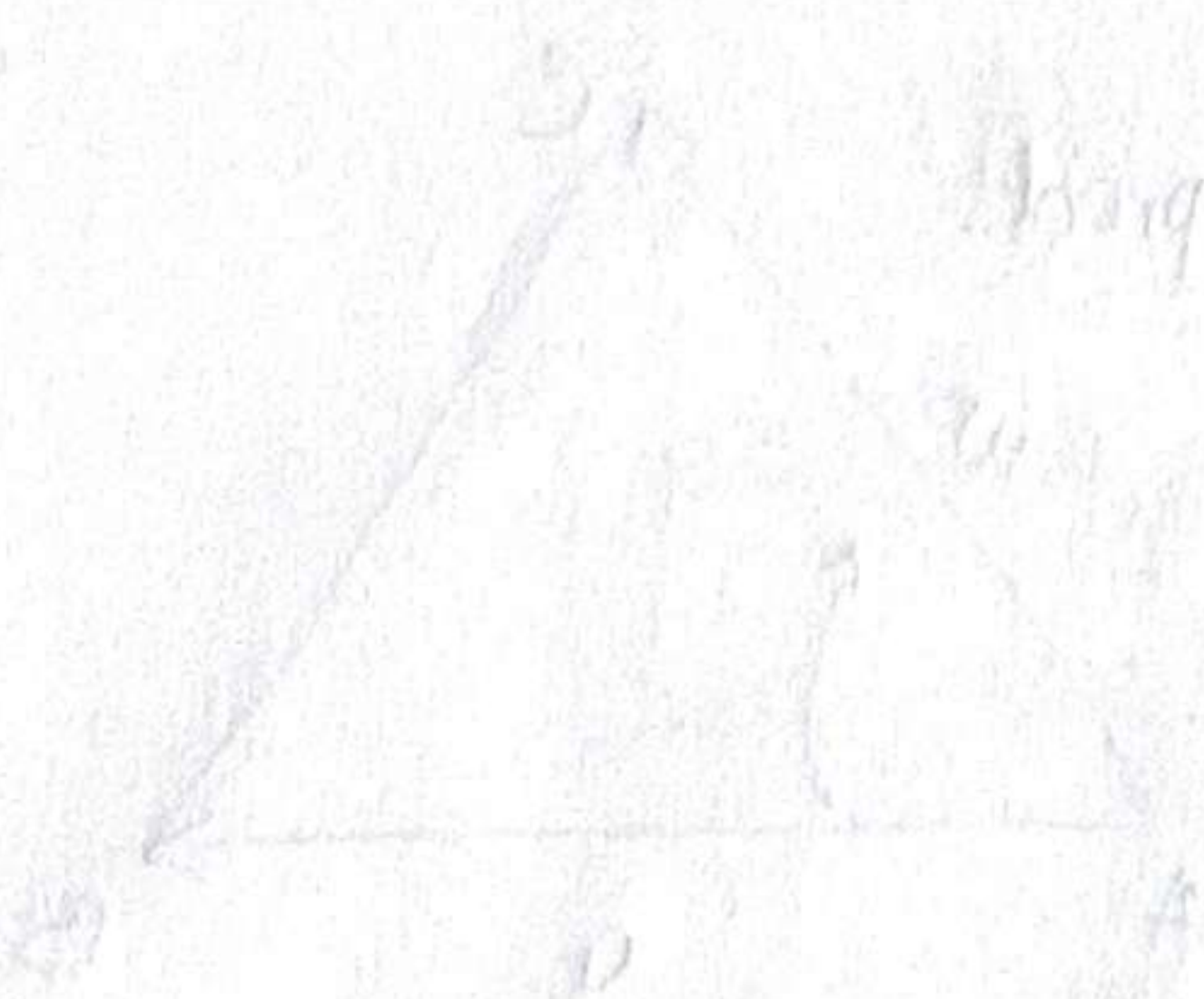
$$\frac{\quad}{(v_1^2 + v_2^2)(w_1^2 + w_2^2)} \quad (+)$$

? **Problem 20** Cross product $\rightarrow$ bilinear
 $\rightarrow$ skew-symmetric.

$\sim$ a, b vectors $\Rightarrow a \times b$ vector.

$$a \times b = |a| \cdot \hat{y}^\perp (P_{r^\perp a}(b))$$

$\hat{y}^\perp a (P_{r^\perp a}(b))$ orthogonal



$\hat{y}^\perp a = \frac{a \times b}{|a| |b|}$

$\hat{y}^\perp a = \frac{a \times b}{|a| |b|}$

$$a \times b = |a| |b| \sin \theta \hat{y}^\perp a$$

$$a \times b = |a| |b| \sin \theta \hat{y}^\perp a$$

Problem 2.1

Jacobi identity for the cross product.

$$(a \times b) \times c + (b \times c) \times a + (c \times a) \times b = 0, \quad \forall a, b, c \in V^3.$$

$$(x \times y) \times z = \langle x, z \rangle y - \langle y, z \rangle x.$$

$$(a \times b) \times c = \langle a, c \rangle b - \langle b, c \rangle a$$

$$(b \times c) \times a = \langle b, a \rangle c - \langle c, a \rangle b$$

$$(c \times a) \times b = \langle c, b \rangle a - \langle a, b \rangle c$$

(+)

$$= 0.$$

Problem 2.2

Prove Lagrange identity.

$$\langle a \times b, c \times d \rangle = \langle a, c \rangle \cdot \langle b, d \rangle - \langle b, c \rangle \cdot \langle a, d \rangle = \begin{vmatrix} \langle a, c \rangle & \langle a, d \rangle \\ \langle b, c \rangle & \langle b, d \rangle \end{vmatrix}$$

Double cross formula: $(x \times y) \times z = \langle x, z \rangle y - \langle y, z \rangle x$

$$x = c$$

$$y = d$$

$$z = a \times b$$

$$\Rightarrow (c \times d)(a \times b) = \langle c, a \times b \rangle d - \langle d, a \times b \rangle c$$

$$(c \times d)(a \times b) \perp a \times b$$

$$\Rightarrow (a \times b) \times d = \langle a, d \rangle b - \langle b, d \rangle a$$

$$\langle a \times b, c \times d \rangle = \langle (a \times b) \times d, c \rangle$$

$$= \langle \langle a, d \rangle b - \langle b, d \rangle a, c \rangle$$

bilinearity $\langle a, d \rangle \langle b, c \rangle - \langle b, d \rangle \langle a, c \rangle$

reorder $\langle a, c \rangle \langle b, d \rangle - \langle b, c \rangle \langle a, d \rangle$

$$= \begin{vmatrix} \langle a, c \rangle & \langle a, d \rangle \\ \langle b, c \rangle & \langle b, d \rangle \end{vmatrix}$$

Problem 23 Prove:

$$(a \times b) \times (c \times d) = b \cdot [a, c, d] - a \cdot [b, c, d] = c \cdot [a, b, d] - d \cdot [a, b, c], \quad (\forall) a, b, c, d$$

$$[a, b, c] = \langle a \times b, c \rangle$$

$$\langle c, a \times b \rangle = [c, a, b] = [a, b, c]$$

$$(x \times y) \times z = \langle x, z \rangle y - \langle y, z \rangle x$$

$$\langle d, a \times b \rangle = [d, a, b] = [a, b, d]$$

$$\begin{aligned} x &= a \\ y &= b \\ z &= c \times d \end{aligned} \quad \Rightarrow \quad (a \times b) \times (c \times d) = \underbrace{\langle a, c \times d \rangle}_{} b - \underbrace{\langle b, c \times d \rangle}_{} a$$

$$= [a, c, d] b - [b, c, d] a$$

$$= b [a, c, d] - a [b, c, d]$$

$$\begin{aligned} \Rightarrow (c \times d) \times (a \times b) &= -\langle c, a \times b \rangle d + \langle d, a \times b \rangle c \\ &= -[c, a, b] d + [d, a, b] c \\ &= -d \cdot [c, a, b] + c \cdot [d, a, b] \\ &= c [a, b, d] - d [a, b, c]. \end{aligned}$$

Problem 24 Distance from a point to a line in $\mathbb{E}^3$.

Distance between two parallel lines in $\mathbb{E}^3$.

$$\text{Area parallelogram} = b \cdot h = |v| \cdot d(P, l) = |\vec{AP} \times v|$$

$$\Rightarrow d(P, l) = \frac{|\vec{AP} \times v|}{|v|} \leftarrow \text{norm.}$$

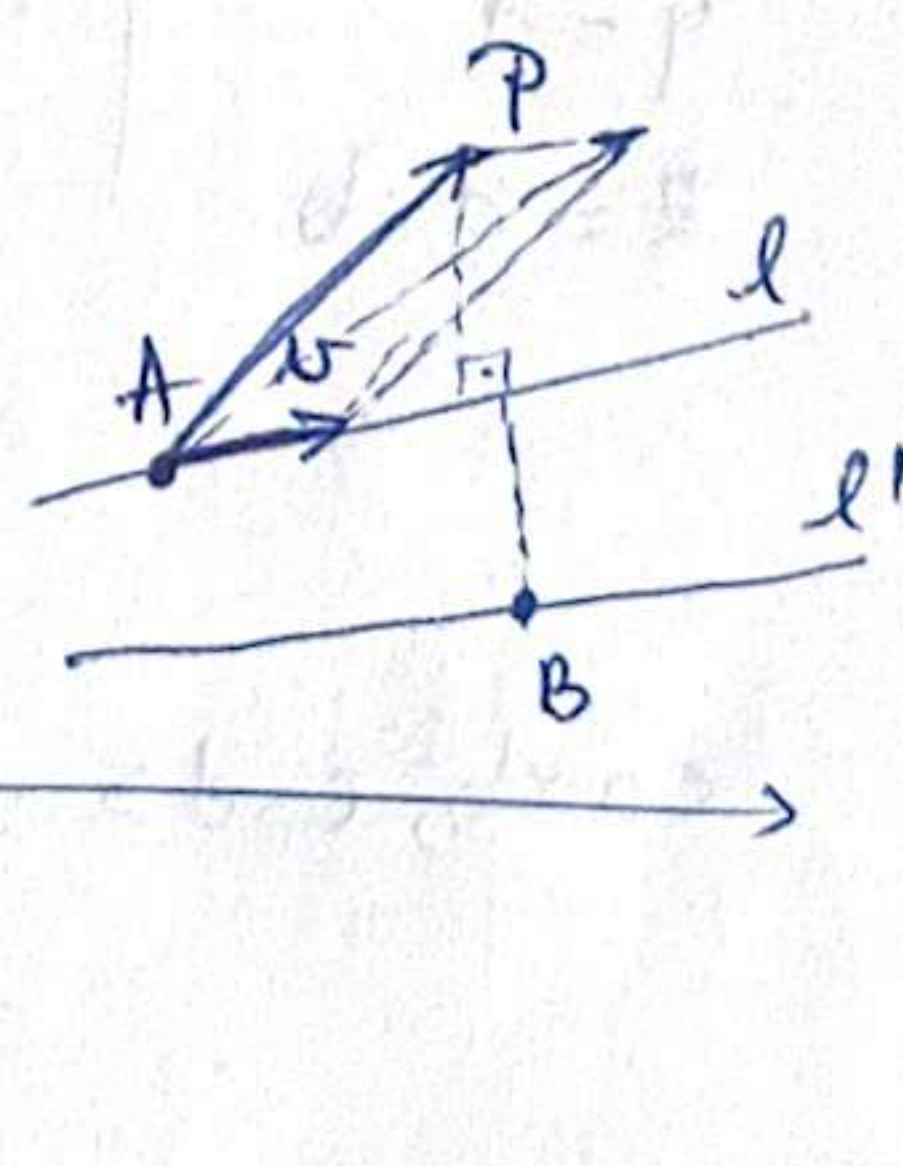
$$A(x_A, y_A, z_A), P(x_P, y_P, z_P), v(v_x, v_y, v_z).$$

$$d(P, l) = \frac{\begin{vmatrix} i & j & k \\ x_P - x_A & y_P - y_A & z_P - z_A \\ v_x & v_y & v_z \end{vmatrix}}{\sqrt{v_x^2 + v_y^2 + v_z^2}}$$

$$|\vec{AP} \times v| = \begin{vmatrix} i & j & k \\ x_P - x_A & y_P - y_A & z_P - z_A \\ v_x & v_y & v_z \end{vmatrix}$$

$$\vec{AP} \times v = \begin{vmatrix} i & j & k \\ x_P - x_A & y_P - y_A & z_P - z_A \\ v_x & v_y & v_z \end{vmatrix}$$

$$l \parallel l'; B \in l' \Rightarrow d(l, l') = d(B, l) = \frac{|\vec{AB} \times v|}{|v|}$$



Ex 25 Eg. for the common $\perp$ line of two skew lines in $\mathbb{E}^3$.

$$l: A + t \cdot v$$

$$v \nparallel w \text{ (skew)} \Rightarrow a = v \times w \neq 0.$$

$$l': B + s \cdot w$$

Take plane $\bar{u}$ through A , parallel to v and a .

$$\Rightarrow l \subset \bar{u} \Rightarrow m = (v \times w) \times v = a \times v$$

(normal)

Take plane $\bar{u}'$ through B , parallel to w and a .

$$\Rightarrow l' \subset \bar{u}' \Rightarrow m' = (v \times w) \times w = a \times w.$$

(normal)

$$d = \bar{u} \cap \bar{u}'$$

$$m \times m' = \left(\overbrace{(v \times w) \times v}^a \right) \times \left(\overbrace{(v \times w) \times w}^a \right) = (a \times v) \times (a \times w)$$

(direction vector of d)

$\downarrow$
proportional to $a = v \times w$. $\Rightarrow$ $\overbrace{a}^a$ is a direction vector of $\underline{d}$.

$$\left. \begin{array}{l} a \perp v \Rightarrow d \perp l \\ a \perp w \Rightarrow d \perp l' \end{array} \right\} \Rightarrow d = l \cap l' \Rightarrow d \text{ is the common } \perp \text{ line.}$$

?

$d = \bar{u} \cap \bar{u}'$

$$A(x_A, y_A, z_A), B(x_B, y_B, z_B), v = (v_x, v_y, v_z), w = (w_x, w_y, w_z).$$

$$a = (a_x, a_y, a_z) = v \times w$$

$$d = \bar{u} \cap \bar{u}'$$

$$\bar{u}: \begin{vmatrix} x - x_A & y - y_A & z - z_A \\ v_x & v_y & v_z \\ a_x & a_y & a_z \end{vmatrix} = 0.$$

$$\bar{u}': \begin{vmatrix} x - x_B & y - y_B & z - z_B \\ w_x & w_y & w_z \\ a_x & a_y & a_z \end{vmatrix} = 0$$

Problem 26 Dim 3; how the box product relates to the cross product and the scalar product. V parallelepiped $\rightarrow$ abs. value of its three edge vectors

• Box prod: $[a, b, c] = \langle a \times b, c \rangle \rightarrow$ scalar triple product.

• $|a \times b| =$ area of parallelogram generated by a, b .

$\Theta \rightarrow$ angle between $a \times b$ and c .

$$\langle a \times b, c \rangle = |a \times b| \cdot |c| \cdot \cos \Theta$$

height

$$V = \underbrace{(\text{base area})}_{|a \times b|} \times \underbrace{(\text{height})}_h = \underbrace{|a \times b|}_b \cdot \underbrace{|c| \cos \Theta}_h = \langle a \times b, c \rangle = [a, b, c]$$

• (a, b, c) x-or $\Rightarrow +$
l-or $\Rightarrow -$

$$\Rightarrow V = |[a, b, c]| = |\langle a \times b, c \rangle|$$

Problem 27 Distance formula between two skew lines in $\mathbb{E}^3$.

$$l: A + t \cdot v$$

$$l': B + s \cdot w$$

$v \nparallel w$ (skew)

• $m = v \times w$ (normal vector)

$$\vec{AB} = B - A$$

$$d(l, l') = \frac{|\langle \vec{AB}, v \times w \rangle|}{|v \times w|} = \frac{|[AB, v, w]|}{|v \times w|} = \frac{\text{BOX pr.}}{\text{CROSS pr.}}$$

$$0 = \begin{vmatrix} AB_x - B_x & AV_x - v_x & AV_y - v_y \\ BV_x & w_x & w_y \\ SV_x & w_x & w_y \end{vmatrix} = 0$$

$$0 = \begin{vmatrix} AB_x - B_x & AV_x - v_x & AV_y - v_y \\ BV_x & w_x & w_y \\ SV_x & w_x & w_y \end{vmatrix} = 0$$